

Ling MAO (毛岭)

Ph.D. Candidate in Integrative Systems and Design
The Hong Kong University of Science and Technology
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Research Interests

Building performance simulation; AI-driven surrogate modeling; building-integrated photovoltaics; climate-resilient building; life-cycle carbon assessment

Education

- *The Hong Kong University of Science and Technology*, 2024.09 – Present
 - ✧ Ph.D. Candidate in Integrative Systems and Design, Cumulative GPA: 3.94/4.00
- *Architectural Association School of Architecture*, 2020.09 – 2022.03
 - ✧ M.Arch in Architecture and Urbanism (DRL), High Pass
- *Zhejiang University*, 2015.09 – 2020.06
 - ✧ B.Arch in Architecture, Cumulative GPA: 3.75/4.00

Peer-Reviewed Journal Articles

- *Mao, L.*, Wu, K., & Xiang, C. (2026). Surrogate-assisted rolling-horizon multi-objective HVAC setpoint optimization for office buildings with BIPV systems, *Energy and Buildings*, 369, 118012.
- *Mao, L.*, & Xiang, C. (2026). Sketch2Radiation: Predicting annual solar radiation maps of buildings from architectural sketches with depth and camera pose, *Building and Environment*, 300, 114725.
- *Mao, L.*, & Xiang, C. (2026). A cross-climate multi-stage approach to life-cycle carbon sensitivity analysis in BIPV-integrated building design, *Energy and Buildings*, 354, 117011.
- Wu, K., *Mao, L.*, & Xiang, C. (2026). Adaptive co-planning of building-integrated photovoltaics and vehicle-to-grid infrastructure under phase-out policy incentives: A two-stage multi-objective optimization framework, *Applied Energy*, 416, 127981.
- *Mao, L.*, & Xiang, C. (2025). Structural colors for building-integrated photovoltaics (BIPV): Multidisciplinary perspectives towards sustainable design, *Materials Today Energy*, 50, 101867.
- *Mao, L.*, & Xiang, C. (2025). A comprehensive review of machine learning applications in perovskite solar cells: Materials discovery, device performance, process optimization and systems integration, *Materials Today Energy*, 47, 101742.

Book Chapters

- Bhooshan, V., Nahmad, A., Singer, P., Chen, T., *Mao, L.*, Louth, H.D. & Bhooshan, S., (2023). Spatial curved laminated timber structures. In *Architecture and Design for Industry 4.0: Theory and Practice* (pp. 859-885). Springer.

Conference Presentations

- **Mao, L.,** Xiang, C. et al., (2026). Occupancy-based climate-responsive control of BIPV smart windows for Hong Kong residential retrofits, *The International Conference on Zero Carbon Built Environment*, Lisbon, Portugal. Oral Presentation.
- **Mao, L.,** & Xiang, C. (2026). Elevated four directional vertical photovoltaic layout strategy for lunar south pole bases, *Academic Forum on Innovation and Development in Extraterrestrial Architecture*, The Architectural Society of China, Harbin, China. Poster Presentation.
- **Mao, L.,** & Xiang, C. (2026). Surrogate-assisted daily rolling multi-objective HVAC setpoint optimization for bipv-integrated office buildings, *The 20th National Environmental Conference for Doctoral Students in Conjunction with 2026 Asian Universities Alliance Postgraduate Academic Forum*, Beijing, China. Oral Presentation.
- **Mao, L.** Yan, Y., & Xiang, C. (2025). Multi-stage and cross-climate sensitivity analysis of design variables in carbon-neutral building design incorporating BIPV, *The International Conference on Carbon Neutrality*, Shanghai, China. Oral Presentation.

Professional Experience

- **Assistant Designer**, Zaha Hadid Architects (扎哈·哈迪德建筑事务所), 2022.04 – 2024.04
 - ✧ *The Greater Bay Area Sports Centre in Nansha* (南沙大湾区文化体育中心, 建成)
 - ✧ *Hangzhou Yuhang International Sports Center* (杭州余杭国际体育中心, 在建)
 - ✧ *Shenzhen Linhai Shuangjie River Bridge* (深圳临海双界河景观桥, 建成)
 - ✧ *Metrotopia Metaverse* (元宇宙项目)
- **Intern**, ISA Internationales Stadtbauatelier (德国 ISA 意厦国际设计集团), 2019.06 – 2019.09
 - ✧ *International Urban Design Competition for Baiyun Lake Digital Technology City*
- **Intern**, The Architectural Design & Research Institute of Zhejiang University, 2018.06 – 2018.09

Awards & Honors

- **Top Cited Paper Award**, Materials Today Energy, Elsevier, 2025
- **RedBird Academic Excellence Award**, Hong Kong University of Science and Technology, 2025
- **HK JEBN Scholarship for Talents**, Hong Kong University of Science and Technology, 2025
- **Finalist Award**, “Guyu Cup” National Sustainable Architecture Design Competition, 2019
- **Academic Excellence Scholarship**, Zhejiang University, 2019
- **Excellent Award**, Zhejiang University SRTP Program, 2018

Technical Skills

- **Programming**: Python, C, C++ , MATLAB, LLM
- **Building Simulation**: EnergyPlus, OpenStudio, Ladybug Tools, CFD, One Click LCA
- **Design and Modeling**: Rhino, Grasshopper, SketchUp, Revit, Unreal Engine, Robotics, Omniverse